NEWTON SCHOOL

ANSWER KEY

Speed, Distance & Time: Word Problems

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Problem 1

Bus route from A to D: 22.5 km + 42.8 km + 42.7 km. Left at 2:30 p.m., arrived at 4:30 p.m. Find average speed.

Total Distance | Time Calculation | Average Speed

Step 1: Find total distance

$\textbf{Total distance} = 22.5 + 42.8 + 42.7 = 108 \text{ km}$

Step 2: Find total time

From 2:30 p.m. to 4:30 p.m. = 2 hours

Step 3: Calculate average speed

$\textbf{Average speed} = \frac{\text{Distance}}{\text{Time}} = \frac{108}{2} = 54 \text{ km/h}$

Answer: 54 km/h

Problem 2

Danny cycled for 2 hours at 10 km/h. (a) Find distance. (b) If speed increased by 2 km/h, find new time.

Distance Formula | Time Formula

(a) Find distance:

$\text{Distance} = \text{Speed} \times \text{Time} = 10 \times 2 = 20 \text{ km}$

(b) Find time with new speed:

New speed = $10 + 2 = 12$ km/h

$\text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{20}{12} = \frac{5}{3} = 1\frac{2}{3} \text{ hours}$

Converting: $1\frac{2}{3}$ hours = 1 hour 40 minutes

(a) 20 km

(b) 1 hour 40 minutes (or $1\frac{2}{3}$ hours)

Problem 3

Ben and David cycled 24 km. David finished 20 min earlier at 9 km/h. Find Ben's speed.

Time Comparison | Speed Calculation

Step 1: Find David's time

$$\text{Time}_{\text{David}} = \frac{24}{9} = \frac{8}{3} \text{ hours} = 2\frac{2}{3} \text{ hours} = 2 \text{ h } 40 \text{ min}$$

Step 2: Find Ben's time

Ben took 20 minutes more than David

$$\text{Time}_{\text{Ben}} = 2 \text{ h } 40 \text{ min} + 20 \text{ min} = 3 \text{ hours}$$

Step 3: Find Ben's speed

$$\text{Speed}_{\text{Ben}} = \frac{24}{3} = 8 \text{ km/h}$$

Answer: 8 km/h

Problem 4

Paul walked 5 min at 46 m/min, then 10 min at 40 m/min. Find total distance.

Distance Formula | Total Distance

Step 1: Distance from house to main road
$\text{Distance}_1 = 46 \times 5 = 230 \text{ m}$

Step 2: Distance from main road to office
$\text{Distance}_2 = 40 \times 10 = 400 \text{ m}$

Step 3: Total distance
$\text{Total} = 230 + 400 = 630 \text{ m}$

Answer: 630 m

Problem 5

12-km hike: first 3 km at 4 km/h, rest at 6 km/h. Find total time.

Time Formula | Total Time

Step 1: Time for first 3 km
$\text{Time}_1 = \frac{3}{4} = 0.75 \text{ hours} = 45 \text{ minutes}$

Step 2: Distance for rest of hike
$\text{Remaining distance} = 12 - 3 = 9 \text{ km}$

Step 3: Time for remaining 9 km
$\text{Time}_2 = \frac{9}{6} = 1.5 \text{ hours} = 1 \text{ h } 30 \text{ min}$

Step 4: Total time
$\text{Total} = 45 \text{ min} + 1 \text{ h } 30 \text{ min} = 2 \text{ h } 15 \text{ min}$

Answer: 2 hours 15 minutes (or 2.25 hours)

Problem 6

Andrew drove: 2 hours for $\frac{4}{5}$ of trip, 1 hour for remaining 30 km. Find (a) total distance, (b) average speed.

Fraction of Distance | Average Speed

(a) Find total distance:

Remaining 30 km = $\frac{1}{5}$ of total distance

If $\frac{1}{5} = 30$ km, then total = $30 \times 5 = 150$ km

(b) Find average speed:

Total distance = 150 km

Total time = 2 + 1 = 3 hours

$$\text{Average speed} = \frac{150}{3} = 50 \text{ km/h}$$

(a) 150 km

(b) 50 km/h

Problem 7

Motorist: $\frac{1}{3}$ of trip at 45 km/h, then 240 km to reach Town Q. (a) Find total distance. (b) If average speed = 54 km/h, find speed for last $\frac{2}{3}$.

Fraction of Distance | Weighted Average Speed

(a) Find total distance:

The 240 km represents $\frac{2}{3}$ of the total trip

If $\frac{2}{3} = 240$ km, then $\frac{1}{3} = 120$ km

Total distance = $120 + 240 = 360$ km

(b) Find speed for last $\frac{2}{3}$:

Step 1: Find total time using average speed

$$\text{Total time} = \frac{360}{54} = \frac{20}{3} \text{ hours}$$

Step 2: Find time for first $\frac{1}{3}$ (120 km at 45 km/h)

$$\text{Time}_1 = \frac{120}{45} = \frac{8}{3} \text{ hours}$$

Step 3: Find time for last $\frac{2}{3}$

$$\text{Time}_2 = \frac{20}{3} - \frac{8}{3} = \frac{12}{3} = 4 \text{ hours}$$

Step 4: Find speed for last $\frac{2}{3}$

$$\text{Speed} = \frac{240}{4} = 60 \text{ km/h}$$

(a) 360 km

(b) 60 km/h

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